

SciOdyssey.

Long-Horizon Autonomous ML Research Agent

Autonomous by default. Supervisable by design.

ROBUST · SUPERVISABLE · BROAD SEARCH

Chen Zhu, Zhou Ziyu, Shilin Xu, GE GAO, Jiran Li



01 | PROBLEM · FAILURE MODES

Why current agents fail

Read the task, inspect data, engineer features, train and evaluate. Reproduce the baseline, then improve it.

01 Closed-world workflow

Predefined tools and recovery paths struggle with unexpected failures.
Design response: an open coding-agent harness that can inspect and repair.

02 Single trajectory

A long-lived context carries prior reasoning into the next experiment.
Design response: fresh Scientist context with independent META review.

03 Tree search

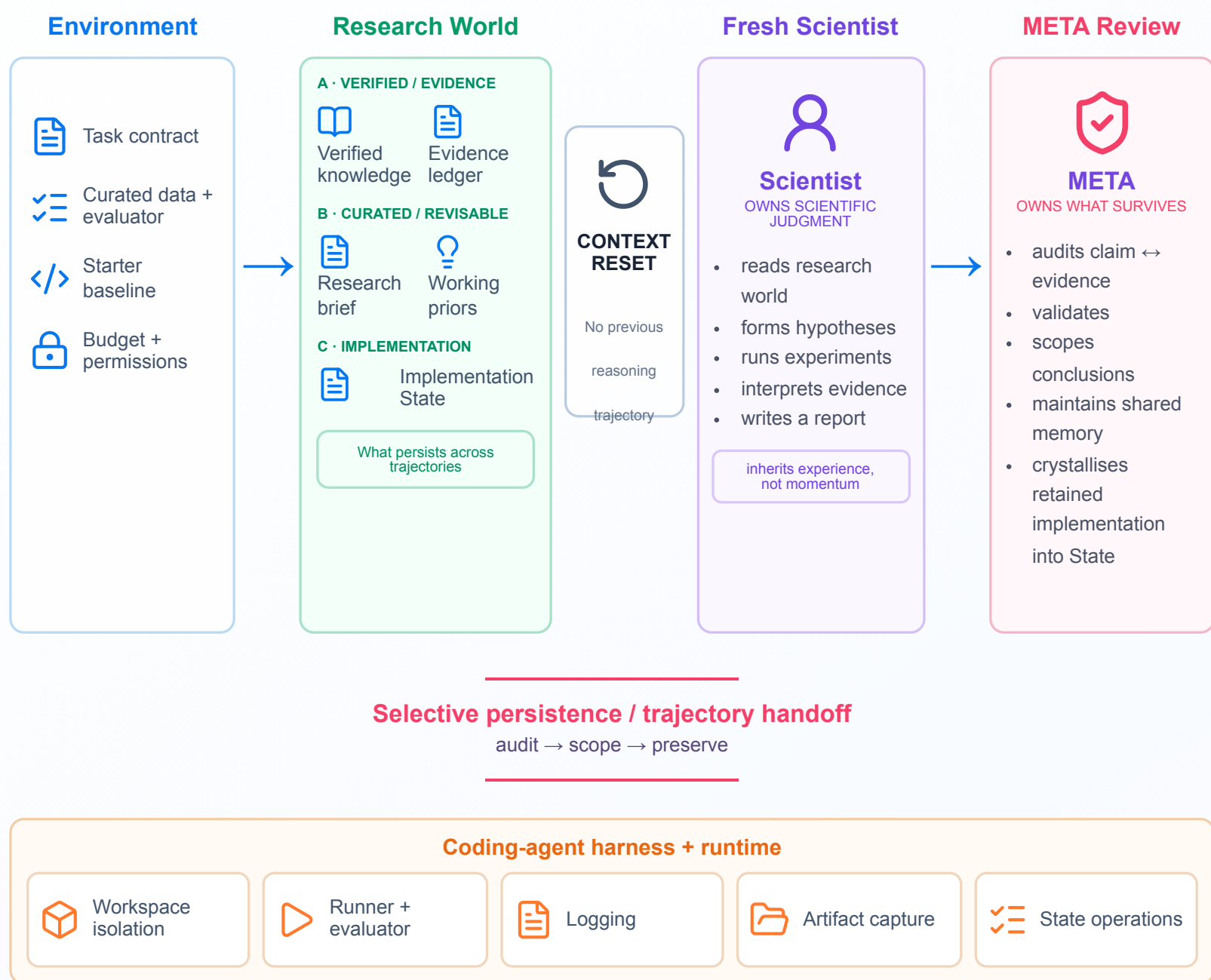
Repeated context reconstruction can cost more than the code edit itself.
One-edit read/write ratio: 8.03×. Eight-child reuse projection: 4.51×.

02 | SYSTEM

SCIENTIST -- META COLLABORATION

A system for sustained research

Give the research world and the researcher different lifetimes.



WORKTREE ISOLATION

Parallel branches explore in separate Git worktrees. Review compares the evidence before an explicit branch promotion.

AUDITED HANDOFF

The next Scientist reads retained evidence, working priors and implementation State. META scopes what survives each trajectory.

04 | EVALUATION

Our agent can carry a research task to completion.

The model got better.

	0.6016000 Official FM baseline	→	0.6059363 SciOdyssey · +0.0043363 Primary
Metric	Official FM	SciOdyssey	Delta
Primary	0.6016000	0.6059363	+0.0043363
GAUC	0.6674000	0.6728421	+0.0054421
nDCG@5	0.5357000	0.5390304	+0.0033304

Primary = mean(GAUC, nDCG@5). Final recipe: 46 categorical feature interactions · 8-seed FM · rank 16 · NumPy / CPU.

The search stayed productive.			
CYCLE 1 0.6016310 Valid baseline 15-field FM	CYCLE 2 0.6040901 Feature expansion 38 fields · 5 seeds	CYCLE 3 0.6044289 Variance reduction 8-seed ensemble	CYCLE 4 0.6059363 Retained recipe 46 interactions

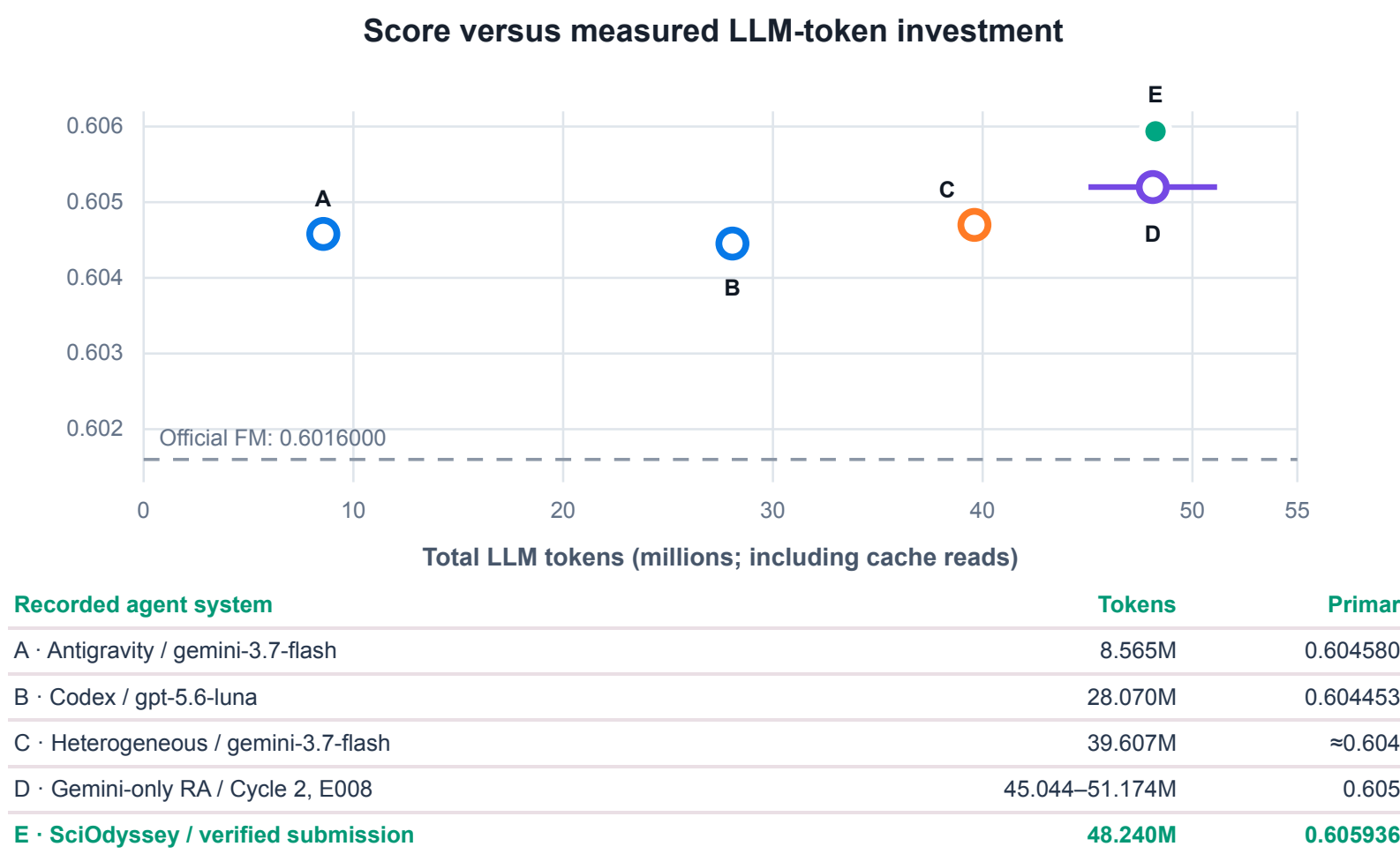
4 autonomous cycles | **13** named experiments | **0** manual scientific interventions*

7 full evaluations · 124,909 validation rows · 22,377 users.
170,588 submission rows passed the Starter Kit alignment check.

* After launch, in the retained submission record.

Compare scores alongside their evidence.

KuaiRand-Pure · public validation · one recorded run per condition



The agent recovered from failure.

Self-healing: In Cycle 1, the Scientist patched numpy.float32 serialization and completed the rerun without human intervention.

META review: In one experiment, sampled training users with unfiltered validation distorted the BPR/BCE ranking. META caught the mismatch and blocked State promotion.

05 | INSIGHTS

01 · MODEL DIVERSITY

Deepen an idea. Open new directions.

In the recorded work, GPT 5.6 Luna refined promising details; Gemini 3.7 Flash explored alternative approaches. The retained Scientist sequence alternated GPT → Gemini → GPT → Gemini, with shared memory maintained by Gemini META.

Gemini-only: **0.6052** · GPT + Gemini: **0.6059363**

Different recipes and stopping boundaries; model rotation was not isolated in an ablation.

02 · PARALLEL RESEARCH

Explore in parallel. Build on the findings.

The reviewer selected branch B (≈0.6048) for further work while retaining modeling ideas from lower-scoring branch C (≈0.6039). A fresh Scientist developed B using C's findings: pairwise ranking, higher rank and rank-normalized ensembling.

Synthesis: **0.6055536** · Public-validation Primary

A lower-scoring branch can still contribute an idea. This reconstructed parallel record is separate from the final submission and does not isolate the effect of parallelism.

03 · ENGINEERING PRACTICE

A clean repo. A continuous agent.

Give every agent a branch. Independent changes keep conflicts local and the main branch clean. **Leave the next move in GitHub.** An issue or prompt can trigger a GitHub Action; review the resulting commit before merging.

Issue / prompt → GitHub Action → Commit → Review

Recommended workflow for unattended exploration, not a measured capability of the retained research run.

RETAINED RUN · 48.240M total tokens · 4.020M non-cache · 0 GPU-hours

References

Repository README; docs/ARCHITECTURE.md;
docs/FINAL_REPORT.md §3.5; public submission package and telemetry.

Contact Information

SciOdyssey Team
github.com/zc6600/research-agent-kuairand